

# IELTS

## Workbook



# Reading

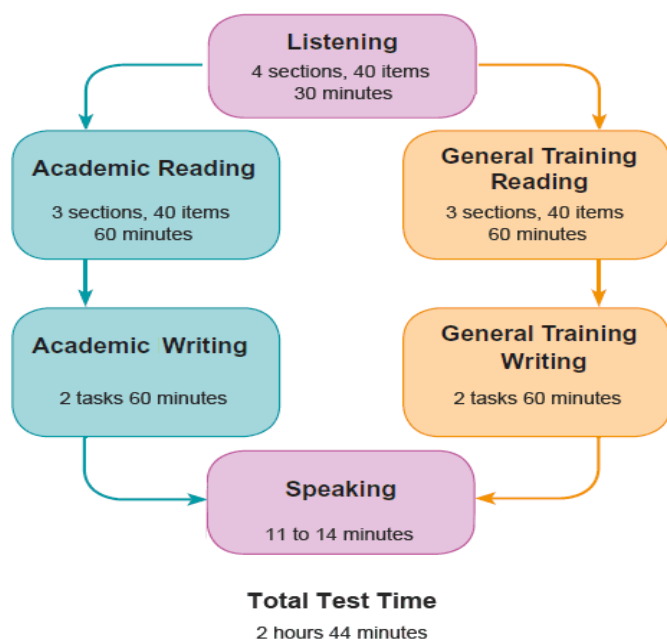
# 8

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## Preface

IELTS (International English Language Testing System) is a worldwiderecognized official English exam. The IELTS is administered by the British Council, IDP: IELTS Australia and Cambridge English Language Assessment. Its aim is to evaluate the English level of the people that take it. This is one of the most accepted English exams by higher education institutions to certify English language proficiency.



**IELTS Reading:** Academic Module – 3 long texts of an academic nature. General Training Module - Mixture of long and short texts of a more general nature, as well as texts related to work situation.

### Reading Paper is Assessed on:

- Identifying the overall purpose
- Follow key arguments in a text
- Identify opinions and attitudes
- Locate specific information
- Distinguish main ideas from details.
- Extract information from a text
- complete a diagram, summary, table

### Reading Types of Questions:

- Matching Heading Questions
- True/ False/Not Given or Yes / No/ Not Given.
- Matching Paragraph Information Question.
- Summary Completion Question
- Sentence Completion Question
- Multiple Choice Question
- List Selection
- Choosing a Title
- Categorization Question
- **Matching Sentence Endings**
- Table Completion
- Flow chart Completion Question
- Short Answer Question

For this type of task, you need to understand how the ideas are linked within the reading passage. This task tests your logical ability as well as your understanding of grammar. The entire passage needs detailed reading and understanding of cohesive devices. There are several ways of linking ideas in a text. Look at the following examples: the findings, as a result..., however..., these questions are commonly asked and account for about 3-4 marks in each passage. Mostly students prefer to solve them later since they need to understand the entire passage well. With our tips & practise papers we are sure you're going to enjoy solving such question and get correct answers.

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## TO SOLVE MATCHING SENTENCE ENDING QUESTIONS:

For this type of task you need to understand how the ideas are linked within the reading passage. This task tests your logical ability as well as your understanding of grammar. The entire passage needs detailed reading and understanding of cohesive devices. There are several ways of linking ideas in a text. Look at the following examples: the findings, as a result..., however..., these questions are commonly asked and account for about 3-4 marks in each passage. Mostly students prefer to solve them later since they need to understand the entire passage well. With our tips & practise papers we are sure you're going to enjoy solving such question and get correct answers

### GUIDELINES TO SOLVE:

1. Skim read the entire passage to get the overall idea of the passage. (5 min)
2. Understand what kind of information is given in each paragraph. You can give headings to each paragraph for your self-understanding. This way it's easy for you to locate the paragraph which contains the information asked and also to link the ideas together.
3. Read the first statement given for the Matching sentence ending questions.
4. Underline the relevant part of the statement to help you locate your answer.
5. Scan the passage and locate the information in the sentence beginnings.
6. The words mentioned in the question are directly taken from the passage so locating them shouldn't be that difficult. They can even use synonyms and paraphrases to test your knowledge of grammar.
7. Carefully read the relevant part of the passage, and then choose the best sentence ending.
8. Spend less than 1 min to analyse each statement.

**Let's look at the application of the above-mentioned guidelines  
for the following passage below.**

## Solved Reading Passage

### COLLECTING ANT SPECIMENS

Collecting ants can be as simple as picking up stray ones and placing them in a glass jar, or as complicated as completing an exhaustive survey of all species present in an area and estimating their relative abundances. The exact method used will depend on the final purpose of the collections. For taxonomy, or classification, long series, from a single nest, which contain all castes (workers, including majors and minors, and, if present, queens and males) are desirable, to allow the determination of variation within species. For ecological studies, the most important factor is collecting identifiable samples of as many of the different species present as possible. Unfortunately, these methods are not always compatible. The taxonomist sometimes overlooks whole species in favour of those groups currently under study, while the ecologist often collects only a limited number of specimens of each species, thus reducing their value for taxonomic investigations.

To collect as wide a range of species as possible, several methods must be used. These include hand collecting, using baits to attract the ants, ground litter sampling, and the use of pitfall traps. Hand collecting consists of searching for ants everywhere they are likely to occur. This includes on the ground, under rocks, logs or other objects on the ground, in rotten wood on the ground or on trees, in vegetation, on tree trunks and under bark. When possible, collections should be made from nests or foraging columns and at least 20 to 25 individuals collected. This will ensure that all individuals are of the same species, and so increase their value for detailed studies. Since some species are largely nocturnal, collecting should not be confined to daytime. Specimens are collected using an aspirator (often called a pooper), forceps, a fine, moistened paint brush, or fingers, if the ants are known not to sting. Individual insects are placed in plastic or glass tubes (1.5-3.0 ml capacity for small ants, 5-8 ml for larger ants) containing 75% to 95% ethanol. Plastic tubes with secure tops are better than glass because they are lighter, and do not break as easily if mishandled.

Baits can be used to attract and concentrate foragers. This often increases the number of individuals collected and attracts species that are otherwise elusive. Sugars and meats or oils will attract different species and a range should be utilised. These baits can be placed either on the ground or on the trunks of trees or large shrubs. When placed on the ground, baits should be situated on small paper cards or other flat, light-coloured surfaces, or in test-tubes or vials. This makes it easier to spot ants and to capture them before they can escape into the surrounding leaf litter.

Many ants are small and forage primarily in the layer of leaves and other debris on the ground. Collecting these species by hand can be difficult. One of the most successful ways to collect them is to gather the leaf litter in which they are foraging and extract the ants from it. This is most commonly done by placing leaf litter on a screen over a large funnel, often under some heat. As the leaf litter dries from above, ants (and other animals) move downward and eventually

fall out the bottom and are collected in alcohol placed below the funnel. This method works especially well in rainforests and marshy areas. A method of improving the catch when using a funnel is to sift the leaf litter through a coarse screen before placing it above the funnel. This will concentrate the litter and remove larger leaves and twigs. It will also allow more litter to be sampled when using a limited number of funnels.

The pitfall trap is another commonly used tool for collecting ants. A pitfall trap can be any small container placed on the ground with the top level with the surrounding surface and filled with a preservative. Ants are collected when they fall into the trap while foraging. The diameter of the traps can vary from about 18 mm to 10 cm and the number used can vary from a few to several hundred. The size of the traps used is influenced largely by personal preference (although larger sizes are generally better), while the number will be determined by the study being undertaken. The preservative used is usually ethylene glycol or propylene glycol, as alcohol will evaporate quickly, and the traps will dry out. One advantage of pitfall traps is that they can be used to collect over a period with minimal maintenance and intervention. One disadvantage is that some species are not collected as they either avoid the traps or do not commonly encounter them while foraging.

Complete each sentence with the correct ending, A-I, below.

Write the correct letter, A-I. **The TWO questions below are solved for you.**

1. Disapene scale insects feed on
2. Neodunetia sangawani ate
3. Leaf-eating hispidies blighted
4. An Argentinian weevil may be successful in wiping out
5. Salvinia molesta plagues

- A. Forage grass
- B. Rice fields
- C. Coconut trees
- D. Fruit trees
- E. Water hyacinth
- F. Parthenium weed
- G. Brazilian beetles
- H. Grass-scale insects
- I. Larval parasites

## Solved Examples:

### 1. Disapene scale insects feed on \_\_\_\_\_.

Disapene scale insects are mentioned in **Paragraph I** which talks about the use of biological control by CIBC on parasites. The answer can be traced from the following sentence, ‘CIBC is also perfecting the technique for breeding parasites that prey on ‘disapene scale’ insects – notorious **defoliants of fruits trees** in the US and India.’ Here the synonym used for the word defoliants in the question is to feed on. This indicates that the correct sentence ending for the above statement is **fruit trees** that is **Option D**

### 2. Neodunetia sangawani ate \_\_\_\_\_

Neodunetia sangawani insect is mentioned in **Paragraph J**. The answer can be traced from the following sentence, ‘A natural predator indigenous to India, Neodunetia sangawani, was found useful in controlling Rhodes **grass-scale insect** that was devouring forage grass in many parts of the US.’ The parasite was useful in controlling Rhodes grass-scale insect means that the parasite feeds on it. This indicates that the correct sentence ending for the above statement is **grass-scale insect** that is **Option H**

## Practise Reading Passage 1.1

### FIERCE, FABULOUS AND FANTASTIC

A new exhibition traces the history of animal painting in Europe from the anatomically inaccurate to the highly sentimental.

The first picture you see in the exhibition *Fierce Friends: Artists and Animals 1750-1900* is of a giraffe – sort of. Painted in about 1785, the creature in it has the neck of a giraffe, but its back is too long, its haunches too developed, and its legs are out of proportion to its body. Like most Europeans in the 18th century, the anonymous French artist who painted it had never seen a real giraffe. He relied on eyewitness descriptions, and on the skin of a giraffe the scientist and adventurer François Levallard had recently brought back from South Africa.

Exotic animals shipped back to Europe at this time usually died soon after arrival, even supposing they survived the voyage. Until about 1900, taxidermy consisted of stuffing the carcass with straw, so the results fell apart after a few years. This meant that ordinary men and women had very few opportunities to see exotic animals at first hand until the establishment of the first zoos – in Paris in 1793, in London in 1818. For an accurate depiction of a giraffe, Europeans had to wait until 1827 and the arrival of the first living specimen, when the Swiss artist Jacques-Laurent Agasse painted his lovely study of the Nubian giraffe sent to King George IV by the Ottoman Viceroy of Egypt.

For most people in the 18th century, animals meant farm animals, carriage horses, and food for the table. But the Enlightenment was an age both of exploration and of discovery, as more and more species of animals, birds, fish and insects were identified and brought back from the South Seas, Africa and India. In 1740, almost 600 species of animals were known to science. One hundred years later, the number had risen to 2,400, including many that are familiar to most children today as a matter of course – ostrich, rhino, orang-utan and buffalo.

Kings and princes, to be sure, had their own menageries, and wealthy collectors added rare birds, fish and mammals (shown side-by-side with two-headed calves and fake dragons) to their cabinets of curiosities. In this way, the forerunners of modern zoos and museums developed along parallel lines. On special occasions, an entrepreneur might exhibit a wild beast to the paying public, as was the case when the Venetian artist Pietro Longhi painted bored masqueraders at carnival time gawping at a pathetic rhinoceros. Out of such displays came another invention of the 19th century, the circus.

Wider knowledge of the animal kingdom came with the publication of George-Louis Leclerc Comte de Buffon's multi-volume *Histoire Naturelle* (1749-88). Based on specimens studied in the royal menageries, this remarkable book is still treasured – not for its scientific accuracy, but for its glorious hand-coloured engravings. Far too expensive for most people to buy, it at least helped to make men and women aware of the beauty of certain animals, as we can see in a

service of Sèvres porcelain created in 1793, where the decorative motifs are taken from the birds drawn by de Buffon.

Gradually, humans began to notice that dumb creatures have feelings. Man cannot afford to feel pity for an animal bred for food. When that wonderful artist Jean-Baptiste Oudry shows a display of dead game in the 1740s, he is simply painting a luxury – fresh meat – available only to the well-off. Peasants ate bread. His lavish paintings were considered suitable for the dining rooms of the nobility because no one then expressed the slightest ethical or moral hesitation about hunting and killing the rabbit, deer and boar for the table, or about slaughtering such vermin as foxes and wolves.

Domestic animals were a different story. When Oudry depicts a hound with her newborn puppies, the simple picture has revolutionary undertones. The pretty white bitch, noticing that two of her pups have fallen asleep and are not getting the nourishment they need, is full of maternal solicitude. At a time when French noblewomen still sent their babies out to wet-nurses, even an animal is shown to display true maternal feeling. And in 1824, the year Delacroix shows two horses killed in battle, there is a new element in man's attitude towards the wanton slaughter of beautiful creatures: compassion. Delacroix's little masterpiece pierces the heart, whereas the grotesque memorial to animals killed in war unveiled in London recently leaves the viewer cold. But the moral impulse behind the creation of both works is exactly the same.

Once animals can be loved for their innocence or good nature, it becomes more difficult to treat them cruelly. Almost 15 years before Jean-Baptiste Greuze painted a picture of a young girl mourning her pet sparrow (1765), William Hogarth published his series of prints, the Stages of Cruelty, showing how the mistreatment of animals leads inexorably to the devaluing of all forms of life, including human. In this show, it is almost impossible to look at Emile Edouard Mouchy's horrifying depiction of the vivisection of a dog (1832) without wincing. Though such experiments represent a necessary evil, our very squeamishness represents another rung upward in the moral evolution of mankind.

This process started in the early 19th century when men began to see in the animal kingdom a mirror image of their own feelings. In his portrayal of a horse frightened by lightning, Gericault lets us see the animal's tensed body, foam-flecked mouth and brow furrowed in anxiety. In The Jealous Lioness of about 1880, the German artist Paul Meyerheim shows a caged lioness enraged at the attention her mate is paying to a beautiful lion tamer.

Gradually, artists began to blur the distinctions between animal and human. When Edwin Landseer in High Life and Low Life contrasts a mongrel guard dog with a deerhound, the animals are surrogates for their absent masters, a butcher and a nobleman. All these artists emphasised the physical and emotional resemblances between animals and human beings.

### Questions 1-4

Complete each sentence with the correct ending, **A-F** below.

Write the correct letter, **A-F** in boxes 1-4 on your answer sheet.

1. Delacroix's 1824 painting \_\_\_\_
2. Greuze's 1765 painting \_\_\_\_
3. Hogarth's series of prints \_\_\_\_
4. Landseer's pair of paintings High Life and Low Life \_\_\_\_

A: makes a moral point about human behaviour.

B: contrasts animal behaviour with human behaviour.

C: shows a human's feeling for a creature.

D: has an identical purpose to that of another work of art.

E: depicts similarities between creatures and people.

F: portrays the feelings creatures can have towards humans.

## Practise Reading Passage 1.2

### NUMERATION

One of the first great intellectual feats of a young child is learning how to talk, closely followed by learning how to count. From earliest childhood, we are so bound up with our system of numeration that it is a feat of imagination to consider the problems faced by early humans who had not yet developed this facility. Careful consideration of our system of numeration leads to the conviction that, rather than being a facility that comes naturally to a person, it is one of the great and remarkable achievements of the human race.

It is impossible to learn the sequence of events that led to our developing the concept of number. Even the earliest of tribes had a system of numeration that, if not advanced, was sufficient for the tasks that they had to perform. Our ancestors had little use for actual numbers; instead, their considerations would have been more of the kind *Is this enough?* rather than *How many?* when they were engaged in food gathering, for example. However, when early humans first began to reflect on the nature of things around them, they discovered that they needed an idea of number simply to keep their thoughts in order. As they began to settle, grow plants and herd animals, the need for a sophisticated number system became paramount. It will never be known how and when this numeration ability developed, but it is certain that numeration was well developed by the time humans had formed even semipermanent settlements.

Evidence of early stages of arithmetic and numeration can be readily found. The indigenous peoples of Tasmania were only able to count one, two, many; those of South Africa counted one, two, two and one, two twos, two twos and one, and so on. But in real situations, the number and words are often accompanied by gestures to help resolve any confusion. For example, when using the one, two, many types of system, the word *many* would mean, *Look my hands and see how many fingers I am showing you*. This basic approach is limited in the range of numbers that it can express, but this range will generally suffice when dealing with the simpler aspects of human existence.

The lack of ability of some cultures to deal with large numbers is not really surprising. European languages, when traced back to their earlier version, are very poor in number words and expressions. The ancient Gothic word for ten, *tachund*, is used to express the number 100 as *tachund tachund*. By the seventh century, the word *teon* had become interchangeable with the *tachund* or *hund* of the Anglo-Saxon language, and so 100 was denoted as *hund teontig*, or ten times ten. The average person in the seventh century in Europe was not as familiar with numbers as we are today. In fact, to qualify as a witness in a court law a man had to be able to count to nine!

Perhaps the most fundamental step in developing a sense of number is not the ability to count, but rather to see that a number is really an abstract idea instead of a simple attachment to a group of particular objects. It must have been within the grasp of the earliest humans to conceive that four birds are distinct from two birds; however, it is not an elementary step to associate the

number 4, as connected with four birds, to the number 4, as connected with four rocks. Associating a number as one of the qualities of a specific object is a great hindrance to the development of a true number sense. When the number 4 can be registered in the mind as a specific word, independent of the object being referenced, the individual is ready to take the first step toward the development of a notational system for numbers and, from there, to arithmetic.

Traces of the very first stages in the development of numeration can be seen in several living languages today. The numeration system of the Tsimshian language in British Columbia contains seven distinct sets of words for numbers according to the class of the item being counted: for counting flat objects and animals, for round objects and time, for people, for long objects and trees, for canoes, for measures, and for counting when no particular object is being numerated. It seems that the last is a later development while the first six groups show the relics of an older system. This diversity of number names can also be found in some widely used languages such as Japanese.

Intermixed with the development of a number sense is the development of an ability to count. Counting is not directly related to the formation of a number concept because it is possible to count by matching the items being counted, against a group of pebbles, grains of corn, or the counter's fingers. These aids would have been indispensable to very early people who would have found the process impossible without some form of mechanical aid. Such aids, while different, are still used even by the most educated in today's society due to their convenience. All counting ultimately involves reference to something other than the things being counted. At first, it may have been grains or pebbles but now it is a memorised sequence of words that happen to be the names of the numbers.

### Questions 1-5

Complete each sentence with the correct ending, **A-G**, below.

Write the correct letter, **A-G**, in boxes 1-5 on your answer sheet.

1. A developed system of numbering \_\_\_\_\_
2. An additional hand signal \_\_\_\_\_
3. In seventh-century Europe, the ability to count to a certain number \_\_\_\_\_
4. Thinking about numbers as concepts separate from physical objects \_\_\_\_\_
5. Expressing number differently according to class of item \_\_\_\_\_

### List of statement endings

- A. was necessary in order to fulfil a civic role.
- B. was necessary when people began farming.
- C. was necessary for the development of arithmetic.
- D. persists in all societies.
- E. was used when the range of number words was restricted.
- F. can be traced back to early European languages.
- G. was a characteristic of early numeration systems.

## Practise Reading Passage 1.3

### FINDING THE LOST FREEDOM

1. The private car is assumed to have widened our horizons and increased our mobility. When we consider our children's mobility, they can be driven to more places (and more distant places) than they could visit without access to a motor vehicle. However, allowing our cities to be dominated by cars has progressively eroded children's independent mobility. Children have lost much of their freedom to explore their own neighbourhood or city without adult supervision. In recent surveys, when parents in some cities were asked about their own childhood experiences, the majority remembered having more, or far more, opportunities for going out on their own, compared with their own children today. They had more freedom to explore their own environment.
2. Children's independent access to their local streets may be important for their own personal, mental and psychological development. Allowing them to get to know their own neighbourhood and community gives them a 'sense of place'. This depends on active exploration', which is not provided for when children are passengers in cars. (Such children may see more, but they learn less.) Not only is it important that children be able to get to local play areas by themselves but walking and cycling journeys to school and to other destinations provide genuine play activities in themselves.
3. There are very significant time and money costs for parents associated with transporting their children to school, sport and to other locations. Research in the United Kingdom estimated that this cost, in 1990, was between 10 billion and 20 billion pounds.
4. The reduction in children's freedom may also contribute to a weakening of the sense of local community. As fewer children and adults use the streets as pedestrians, these streets become less sociable places. There is less opportunity for children and adults to have the spontaneous of the community. This in itself may exacerbate fears associated with assault and molestation of children, because there are fewer adults available who know their neighbours' children, and who can look out for their safety.
5. The extra traffic involved in transporting children results in increased traffic congestion, pollution and accident risk. As our roads become more dangerous, more parents drive their children to more places, thus contributing to increased levels of danger for the remaining pedestrians. Anyone who has experienced either the reduced volume of traffic in peak hour during school holidays, or the traffic jams near schools at the end of a school day, will not need convincing about these points. Thus, there are also important environmental implications of children's loss of freedom.
6. As individuals, parents strive to provide the best upbringing they can for their children. However, in doing so, (e.g. by driving their children to sport, school or recreation) parents

may be contributing to a more dangerous environment for children generally. The idea that 'streets are for cars and backyards and playgrounds are for children' is a strongly held belief, and parents have little choice as individuals but to keep their children off the streets if they want to protect their safety.

7. In many parts of Dutch cities, and some traffic calmed precincts in Germany, residential streets are now places where cars must give way to pedestrians. In these areas, residents are accepting the view that the function of streets is not solely to provide mobility for cars. Streets may also be for social interaction, walking, cycling and playing. One of the most important aspects of these European cities, in terms of giving cities back to children, has been a range of 'traffic calming' initiatives, aimed at reducing the volume and speed of traffic. These initiatives have had complex interactive effects, leading to a sense that children have been able to 'recapture' their local neighbourhood, and more importantly, that they have been able to do this in safety. Recent research has demonstrated that children in many German cities have significantly higher levels of freedom to travel to places in their own neighbourhood or city than children in other cities in the world.
8. Modifying cities in order to enhance children's freedom will not only benefit children. Such cities will become more environmentally sustainable, as well as more sociable and more livable for all city residents. Perhaps it will be our concern for our children's welfare that convinces us that we need to challenge the dominance of the car in our cities.

### Questions 1-5

Questions 1-5 are statement beginnings which represent information given in Paragraphs 6, 7 and 8. In the box below, there are some statement endings numbered A-J. Choose the correct ending for each statement.

**Write your answers A-J, in the spaces numbered 1-5 on the answer sheet.**

1. Children should play \_\_\_\_\_
2. In some German towns, pedestrians have right of way \_\_\_\_\_
3. Streets should also be used for \_\_\_\_\_
4. Reducing the amount of traffic and the speed is \_\_\_\_\_
5. All people who live in the city will benefit if cities are \_\_\_\_\_

#### List of statement endings

- |                             |                     |
|-----------------------------|---------------------|
| A. a dangerous environment. | F. Socializing.     |
| B. modified.                | G. in backyards.    |
| C. on residential streets.  | H. for cars         |
| D. modifying cities.        | I. traffic calming. |
| E. neighbourhoods.          | J. residential      |

## Practise Reading Passage 1.4

### UNMASKING SKIN

- A. If you took off your skin and laid it flat, it would cover an area of about twenty-one square feet, making it by far the body's largest organ. Draped in place over our bodies, skin forms the barrier between what's inside us and what's outside. It protects us from a multitude of external forces. It serves as an avenue to our most intimate physical and psychological selves.
- B. This impervious yet permeable barrier, less than a millimetre thick in places, is composed of three layers. The outermost layer is the bloodless epidermis. The dermis includes collagen, elastin, and nerve endings. The innermost layer, subcutaneous fat, contains tissue that acts as an energy source, cushion and insulator for the body.
- C. From these familiar characteristics of skin emerge the profound mysteries of touch, arguably our most essential source of sensory stimulation. We can live without seeing or hearing – in fact, without any of our other senses. But babies born without effective nerve connections between skin and brain can fail to thrive and may even die.
- D. Laboratory experiments decades ago, now considered unethical and inhumane, kept baby monkeys from being touched by their mothers. It made no difference that the babies could see, hear and smell their mothers; without touching, the babies became apathetic, and failed to progress.
- E. For humans, insufficient touching in early years can have lifelong results. "In touching cultures, adult aggression is low, whereas, in cultures where touch is limited, adult aggression is high," writes Tiffany Field, director of the Touch Research Institutes at the University of Miami School of Medicine. Studies of a variety of cultures show a correspondence between high rates of physical affection in childhood and low rates of adult physical violence.
- F. While the effects of touching are easy to understand, the mechanics of it are less so. "Your skin has millions of nerve cells of various shapes at different depths," explains Stanley Bolanowski, a neuroscientist and associate director of the Institute for Sensory Research at Syracuse University. "When the nerve cells are stimulated, physical energy is transformed into energy used by the nervous system and passed from the skin to the spinal cord and brain. It's called transduction, and no one knows exactly how it takes place." Suffice it to say that the process involves the intricate, split-second operation of a complex system of signals between neurons in the skin and brain.
- G. This is starting to sound very confusing until Bolanowski says: "In simple terms, people perceive three basic things via skin: pressure, temperature, and pain." And then I'm sure he's

wrong. "When I get wet, my skin feels wet," I protest. "Close your eyes and lean back," says Bolanowski.

- H. Something cold and wet is on my forehead – so wet, in fact, that I wait for water to start dripping down my cheeks. "Open your eyes." Bolanowski says, showing me that the sensation comes from a chilled, but dry, metal cylinder. The combination of pressure and cold, he explains, is what makes my skin perceive wetness. He gives me a surgical glove to put on and has me put a finger in a glass of cold water. My finger feels wet, even though I have visual proof that it's not touching water. My skin, which seemed so reliable, has been deceiving me my entire life. When I shower or wash my hands, I now realize, my skin feels pressure and temperature. It's my brain that says I feel wet.
- I. Perceptions of pressure, temperature and pain manifest themselves in many different ways. Gentle stimulation of pressure receptors can result in ticklishness; gentle stimulation of pain receptors, in itching. Both sensations arise from a neurological transmission, not from something that physically exists. Skin, I'm realizing, is under constant assault, both from within the body and from forces outside. Repairs occur with varying success.
- J. Take the spot where I nicked myself with a knife while slicing fruit. I have a crusty scab surrounded by pink tissue about a quarter inch long on my right palm. Under the scab, epidermal cells are migrating into the wound to close it up. When the process is complete, the scab will fall off to reveal new epidermis. It's only been a few days, but my little self-repair is almost complete. Likewise, we recover quickly from slight burns. If you ever happen to touch a hot burner, just put your finger in cold water. The chances are you will have no blister, little pain and no scar. Severe burns, though, are a different matter.

### Questions 1-5

Complete each sentence with the correct ending **A–I** from the box below. Write the correct letter **A–I** in boxes 1-5 on your answer sheet.

1. Touch is unique among the five senses \_\_\_\_\_
2. A substance may feel wet \_\_\_\_\_
3. Something may tickle \_\_\_\_\_
4. The skin may itch \_\_\_\_\_
5. A small cut heals up quickly \_\_\_\_\_

- A. because it is both cold and painful
- B. because the outer layer of the skin can mend itself.
- C. because it can be extremely thin.
- D. because there is light pressure on the skin.
- E. because we do not need the others to survive.
- F. because there is a good blood supply to the skin.
- G. because of a small amount of pain.
- H. because there is a low temperature and pressure.
- I. because it is hurting a lot.
- J. because all humans are capable of experiencing it.

## Practise Reading Passage 1.5

### THE RETURN OF ARTIFICIAL INTELLIGENCE

*It is becoming acceptable again to talk of computers performing human tasks such as problem-solving and pattern-recognition.*

- A. After years in the wilderness, the term 'artificial intelligence' (AI) seems poised to make a comeback. AI was big in the 1980s but vanished in the 1990s. It re-entered public consciousness with the release of AI, a movie about a robot boy. This has ignited a public debate about AI, but the term is also being used once more within the computer industry. Researchers, executives and marketing people are now using the expression without irony or inverted commas. And it is not always hype. The term is being applied, with some justification, to products that depend on technology that was originally developed by AI researchers. Admittedly, the rehabilitation of the term has a long way to go, and some firms still prefer to avoid using it. But the fact that others are starting to use it again suggests that AI has moved on from being seen as an over-ambitious and under-achieving field of research.
- B. The field was launched, and the term 'artificial intelligence' coined, at a conference in 1956 by a group of researchers that included Marvin Minsky, John McCarthy, Herbert Simon and Alan Newell, all of whom went on to become leading figures in the field. The expression provided an attractive but informative name for a research programme that encompassed such previously disparate fields as operations research, cybernetics, logic and computer science. The goal they shared was an attempt to capture or mimic human abilities using machines. That said, different groups of researchers attacked different problems, from speech recognition to chess playing, in different ways; AI unified the field in name only. But it was a term that captured the public imagination.
- C. Most researchers agree that AI peaked around 1985. A public reared on science-fiction movies and excited by the growing power of computers had high expectations. For years, AI researchers had implied that a breakthrough was just around the corner. Marvin Minsky said in 1967 that within a generation the problem of creating 'artificial intelligence' would be substantially solved. Prototypes of medical-diagnosis programs and speech recognition software appeared to be making progress. It proved to be a false dawn. Thinking computers and household robots failed to materialise, and a backlash ensued. 'There was undue optimism in the early 1980s; says David Leaky, a researcher at Indiana University. 'Then when people realised these were hard problems, there was retrenchment. By the late 1980s, the term AI was being avoided by many researchers, who opted instead to align themselves

with specific sub-disciplines such as neural networks, agent technology, case-based reasoning, and so on.

- D. Ironically, in some ways AI was a victim of its own success. Whenever an apparently mundane problem was solved, such as building a system that could land an aircraft unattended, the problem was deemed not to have been AI in the first place. 'If it works, it can't be AI; as Dr Leaky characterises it. The effect of repeatedly moving the goal-posts in this way was that AI came to refer to 'blue-sky' research that was still years away from commercialisation. Researchers joked that AI stood for 'almost implemented'. Meanwhile, the technologies that made it onto the market, such as speech recognition, language translation and decision-support software, were no longer regarded as AI. Yet all three once fell well within the umbrella of AI research.
- E. But the tide may now be turning, according to Dr Leake. HNC Software of San Diego, backed by a government agency, reckons that their new approach to artificial intelligence is the most powerful and promising approach ever discovered. HNC claim that their system, based on a cluster of 30 processors, could be used to spot camouflaged vehicles on a battlefield or extract a voice signal from a noisy background - tasks humans can do well, but computers cannot. 'Whether or not their technology lives up to the claims made for it, the fact that HNC are emphasising the use of AI is itself an interesting development; says Dr Leaky.
- F. Another factor that may boost the prospects for AI in the near future is that investors are now looking for firms using clever technology, rather than just a clever business model, to differentiate themselves. In particular, the problem of information overload, exacerbated by the growth of e-mail and the explosion in the number of web pages, means there are plenty of opportunities for new technologies to help filter and categorise information - classic AI problems. That may mean that more artificial intelligence companies will start to emerge to meet this challenge.
- G. The 1969 film, 2001: A Space Odyssey, featured an intelligent computer called HAL 9000. As well as understanding and speaking English, HAL could play chess and even learned to lip-read. HAL thus encapsulated the optimism of the 1960s that intelligent computers would be widespread by 2001. But 2001 has been and gone, and there is still no sign of a HAL-like computer. Individual systems can play chess or transcribe speech, but a general theory of machine intelligence still remains elusive. It may be, however, that the comparison with

HAL no longer seems quite so important, and AI can now be judged by what it can do, rather than by how well it matches up to a 30-year-old science-fiction film. 'People are beginning to realise that there are impressive things that these systems can do; says Dr Leake hopefully.

### Question 1-5

Complete each sentence with the correct ending, A-G, below. Write the correct letter, A-G.

1. In future, investors are looking forward to invest in AI companies using
2. Present day research on AI does not compare its results with
3. Initially, speech recognition, language transition and decision support software
4. The main goal of the Research group was
5. One of the main reasons for the comeback of AI in 1990's

- A. Were parts of the AI research.
- B. Clever technology
- C. Release of a movie based on a robot.
- D. Develop machines that can imitate human abilities.
- E. 2001: A space Odyssey film.
- F. Were not parts of the AI research.
- G. HAL-9000

## Practise Reading Passage 1.6

### ALARMING RATE OF LOSS OF TROPICAL RAINFOREST

Adults and children are frequently confronted with statements about the alarming rate of loss of tropical rainforests.

For example, one graphic illustration to which children might readily relate is the estimate that rainforests are being destroyed at a rate equivalent to one thousand football fields every forty minutes – about the duration of a normal classroom period. In the face of the frequent and often vivid media coverage, it is likely that children will have formed ideas about rainforests – what and where they are, why they are important, what endangers them – independent of any formal tuition. It is also possible that some of these ideas will be mistaken. Many studies have shown that children harbour misconceptions about ‘pure’, curriculum science. These misconceptions do not remain isolated but become incorporated into a multifaceted, but organised, conceptual framework, making it and the component ideas, some of which are erroneous, more robust but also accessible to modification. These ideas may be developed by children absorbing ideas through the popular media. Sometimes this information may be erroneous. It seems schools may not be providing an opportunity for children to re-express their ideas and so have them tested and refined by teachers and their peers.

Despite the extensive coverage in the popular media of the destruction of rainforests, little formal information is available about children’s ideas in this area. The aim of the present study is to start to provide such information, to help teachers design their educational strategies to build upon correct ideas and to displace misconceptions and to plan programmes in environmental studies in their schools.

The study surveys children’s scientific knowledge and attitudes to rainforests. Secondary school children were asked to complete a questionnaire containing five open-form questions. The most frequent responses to the first question were descriptions which are self-evident from the term ‘rainforest’. Some children described them as damp, wet or hot. The second question concerned the geographical location of rainforests. The commonest responses were continents or countries: Africa (given by 43% of children), South America (30%), Brazil (25%). Some children also gave more general locations, such as being near the Equator.

Responses to question three concerned the importance of rainforests. The dominant idea, raised by 64% of the pupils, was that rainforests provide animals with habitats. Fewer students responded that rainforests provide plant habitats, and even fewer mentioned the indigenous populations of rainforests. More girls (70%) than boys (60%) raised the idea of the rainforest as animal habitats.

Similarly, but at a lower level, more girls (13%) than boys (5%) said that rainforests provided human habitats. These observations are generally consistent with our previous studies of pupils’ views about the use and conservation of rainforests, in which girls were shown to be more

sympathetic to animals and expressed views which seem to place an intrinsic value on non-human animal life.

The fourth question concerned the causes of the destruction of rainforests. Perhaps encouragingly, more than half of the pupils (59%) identified that it is human activities which are destroying rainforests, some personalising the responsibility by the use of terms such as ‘we are’. About 18% of the pupils referred specifically to logging activity.

One misconception, expressed by some 10% of the pupils, was that acid rain is responsible for rainforest destruction; a similar proportion said that pollution is destroying rainforests. Here, children are confusing rainforest destruction with damage to the forests of Western Europe by these factors. While two-fifths of the students provided the information that the rainforests provide oxygen, in some cases this response also embraced the misconception that rainforest destruction would reduce atmospheric oxygen, making the atmosphere incompatible with human life on Earth.

In answer to the final question about the importance of rainforest conservation, the majority of children simply said that we need rainforests to survive. Only a few of the pupils (6%) mentioned that rainforest destruction may contribute to global warming. This is surprising considering the high level of media coverage on this issue. Some children expressed the idea that the conservation of rainforests is not important.

The results of this study suggest that certain ideas predominate in the thinking of children about rainforests. Pupils’ responses indicate some misconceptions in the basic scientific knowledge of rainforest’s ecosystems such as their ideas about rainforests as habitats for animals, plants and humans and the relationship between climatic change and destruction of rainforests. Pupils did not volunteer ideas that suggested that they appreciated the complexity of causes of rainforest destruction. In other words, they gave no indication of an appreciation of either the range of ways in which rainforests are important or the complex social, economic and political factors which drive the activities which are destroying the rainforests. One encouragement is that the results of similar studies about other environmental issues suggest that older children seem to acquire the ability to appreciate, value and evaluate conflicting views. Environmental education offers an arena in which these skills can be developed, which is essential for these children as future decision-makers.

### Questions 1-5

Complete the following sentences using the list of suitable endings given below.

- 1) The children's most frequent response when asked where the rainforests were was  
\_\_\_\_\_.
- 2) The most common response to the question about the importance of the rainforests was  
\_\_\_\_\_.
- 3) The reason given by most of the children for the loss of rainforest was  
\_\_\_\_\_.
- 4) Most of the children thought it was important to protect rainforest as  
\_\_\_\_\_.
- 5) The responses which were cited as unexpectedly uncommon, given the amount of time spent on the issue by the newspapers and television were  
\_\_\_\_\_.

- A. There is a complicated combination of reasons for the loss of the rainforests.
- B. The rainforests are being destroyed by the same things that are destroying the forests of Western Europe.
- C. Rainforests are located near the Equator.
- D. Brazil is home to the rainforests.
- E. Without rainforests some animals would have nowhere to live.
- F. Rainforests are important habitats for a lot of plants.
- G. People are responsible for the loss of the rainforests.
- H. The rainforests are a source of oxygen.
- I. Rainforests are of consequence for a number of different reasons.
- J. As the rainforests are destroyed, the world gets warmer.
- K. Without rainforests there would not be enough oxygen in the air.
- L. There are people for whom the rainforests are home.
- M. Rainforests are found in Africa.
- N. Rainforests are not really important to human life.
- O. The destruction of the rainforests is the direct result of logging activity.
- P. Humans depend on the rainforests for their continuing existence.

## Practise Reading Passage 1.7

### GREYING POPULATION STAYS IN THE PINK

Elderly people are growing healthier, happier and more independent, say American scientists. The results of a 14-year study to be announced later this month reveal that the diseases associated with old age are afflicting fewer and fewer people and when they do strike, it is much later in life.

In the last 14 years, the National Long-term Health Care Survey has gathered data on the health and lifestyles of more than 20,000 men and women over 65. Researchers, now analysing the results of data gathered in 1994, say arthritis, high blood pressure and circulation problems -the major medical complaints in this age group - are troubling a smaller proportion every year. And the data confirms that the rate at which these diseases are declining continues to accelerate. Other diseases of old age - dementia, stroke, arteriosclerosis and emphysema - are also troubling fewer and fewer people.

'It really raises the question of what should be considered normal ageing,' says Kenneth Manton, a demographer from Duke University in North Carolina. He says the problems doctors accepted as normal in a 65-year-old in 1982 are often not appearing until people are 70 or 75.

Clearly, certain diseases are beating a retreat in the face of medical advances. But there may be other contributing factors. Improvements in childhood nutrition in the first quarter of the twentieth century, for example, gave today's elderly people a better start in life than their predecessors.

On the downside, the data also reveals failures in public health that have caused surges in some illnesses. An increase in some cancers and bronchitis may reflect changing smoking habits and poorer air quality, say the researchers. 'These may be subtle influences,' says Manton, 'but our subjects have been exposed to worse and worse pollution for over 60 years. It's not surprising we see some effect.'

One interesting correlation Manton uncovered is that better-educated people are likely to live longer. For example, 65-year-old women with fewer than eight years of schooling are expected, on average, to live to 82. Those who continued their education live an extra seven years. Although some of this can be attributed to a higher income, Manton believes it is mainly because educated people seek more medical attention.

The survey also assessed how independent people over 65 were, and again found a striking trend. Almost 80% of those in the 1994 survey could complete everyday activities ranging from eating and dressing unaided to complex tasks such as cooking and managing their finances. That represents a significant drop in the number of disabled old people in the population. If the trends apparent in the United States 14 years ago had continued, researchers calculate there would be an additional one million disabled elderly people in today's population. According to Manton,

slowing the trend has saved the United States government's Medicare system more than \$200 billion, suggesting that the greying of America's population may prove less of a financial burden than expected.

The increasing self-reliance of many elderly people is probably linked to a massive increase in the use of simple home medical aids. For instance, the use of raised toilet seats has more than doubled since the start of the study, and the use of bath seats has grown by more than 50%. These developments also bring some health benefits, according to a report from the MacArthur Foundation's research group on successful ageing. The group found that those elderly people who were able to retain a sense of independence were more likely to stay healthy in old age.

Maintaining a level of daily physical activity may help mental functioning, says Carl Cotman, a neuroscientist at the University of California at Irvine. He found that rats that exercise on a treadmill have raised levels of brain-derived neurotrophic factor coursing through their brains. Cotman believes this hormone, which keeps neurons functioning, may prevent the brains of active humans from deteriorating.

As part of the same study, Teresa Seeman, a social epidemiologist at the University of Southern California in Los Angeles, found a connection between self-esteem and stress in people over 70. In laboratory simulations of challenging activities such as driving, those who felt in control of their lives pumped out lower levels of stress hormones such as cortisol. Chronically high levels of these hormones have been linked to heart disease.

But independence can have drawbacks. Seeman found that elderly people who felt emotionally isolated maintained higher levels of stress hormones even when asleep. The research suggests that older people fare best when they feel independent but know they can get help when they need it.

'Like much research into ageing, these results support common sense,' says Seeman. They also show that we may be underestimating the impact of these simple factors. 'The sort of thing that your grandmother always told you turns out to be right on target,' she says.

### Questions 1-4

Complete each sentence with the correct ending, **A-H**, below. Write the correct letter, **A-H**

1. Home medical aids
2. Regular amounts or exercise
3. Feelings of control over life
4. Feelings of loneliness

- A. may cause heart disease.
  - B. can be helped by hormone treatment.
  - C. may cause rises in levels of stress hormones.
  - D. have cost the United States government more than \$200 billion.
  - E. may help prevent mental decline.
  - F. may get stronger at night.
  - G. allow old people to be more independent.
  - H. can reduce stress in difficult situations.

## Practise Reading Passage 1.8

### MAKETE INTEGRATED RURAL TRANSPORT PROJECT

#### Section A

The disappointing results of many conventional road transport projects in Africa led some experts to rethink the strategy by which rural transport problems were to be tackled at the beginning of the 1980s. A request for help in improving the availability of transport within the remote Makete District of southwestern Tanzania presented the opportunity to try a new approach.

The concept of 'integrated rural transport' was adopted in the task of examining the transport needs of the rural households in the district. The objective was to reduce the time and effort needed to obtain access to essential goods and services -through an improved rural transport system. The underlying assumption was that the time saved would be used instead for activities that would improve the social and economic development of the communities. The Makete Integrated Rural Transport - Project (MIRT P) started in 1985 with financial support from the Swiss Development Corporation and was coordinated with the help of the Tanzanian government.

#### Section B

When the project began Makete District was virtually totally isolated during the rainy season. The regional road was in such bad shape that access to the main towns was impossible for about three months of the year. Road traffic was extremely rare with the district, and alternative means of transport were restricted to donkeys in the north of the district, people relied primarily on the paths, which were supper and dangerous during the rains.

Before solutions could be proposed, the problems had to be understood. Little was known about the transport demands of the rural households, so Phase I, between December 1985 and December 1987, focused on research. The socio-economic survey of more than 400 households in the district indicates that a household in Makete spent, on average, seven hours a day on transporting themselves and their goods, a figure which seemed extreme but which has also been obtained in surveys in other rural areas in Africa. Interesting facts regarding transport were found- 95% was on foot, 80% was within the locality and 70% was related to the collection of water and firewood and travelling to running mills.

#### Section C

Having determined the main transport needs, possible solutions were identified which might reduce the time and burden During Phase II. from January to February 1991, a number of approaches were implemented in an effort to improve mobility and access to transport.

An improvement of the rotted network was considered necessary to ensure the import and export of goods to the district. These improvements were carried out using methods that were heavily dependent on labour. In addition to the improvement of roads, these methods provided training

in the operation of a mechanical workshop and bus and truck services. However, the difference from the conventional approach was that this time consideration was given to local transport needs outside the road network.

Most goods were transported along the paths that provide shortcuts up and down the hillsides, but the paths were a real safety task and made the journey on foot even more arduous. It made sense to improve the paths by building steps, handrails and footbridges.

It was uncommon to fix the means of transport that were more efficient than walking but less technologically advanced than motor vehicles. The use of bicycles was constrained by their high cost and the lack of available spare parts. Oxen were not used at all but donkeys were used by a few households in the northern part of the district. MIRTP focused on what would be most appropriate for the inhabitants of Makete in terms of what was available, how much they could afford and what they are willing to accept. After careful consideration, the project chose the promotion of donkeys - a donkey costs less than a bicycle - and the introduction of a locally manufacturable wheelbarrow.

#### **Section D**

At the end of Phase II, it was clear that the selected approaches to Makete's transport problems had had different degrees of success. Phase III, from March 1991 to March 1993, focused on the refinement and most of these activities.

The road improvements and accompanying maintenance system had helped make the district centre accessible throughout the year. Essential goods from outside the district had become more readily available at the market and prices did not fluctuate as much as they had done before.

Paths and secondary roads were improved only at the request of communists who were willing to participate in construction and maintenance. However, the improved paths impressed the inhabitants, and requests for assistance greatly increased soon after only a few improvements had been completed.

The efforts to improve the efficiency of the existing transport services were not very successful because most of the motorised vehicles in the district broke down and there were no resources to repair them. Even the introduction of low-cost means of transport was difficult because of the general poverty of the district. The locally manufactured wheelbarrows were still too expensive for all but a few of the households. Modifications to the original design by local carpenters cut production time and costs. Other local carpenters have been trained in the new design so that they can respond to requests. Nevertheless, a locally produced wooden wheelbarrow which costs around 500Q Tanzanian shillings (less than US\$20) in Makete, and is about one-quarter the cost of a metal wheelbarrow, is still too expensive for most people.

Donkeys, which were imported to the district have become more common and contribute, in particular, to the transportation of crops and goods to market. Those who have bought donkeys

are mainly from richer households but with an increased supply through local breeding, donkeys should become more affordable. Meanwhile, local initiatives are promoting the renting out of the existing donkeys.

It should be noted, however, that a donkey, which at 20,000 Tanzanian shillings costs less than a bicycle, is still an investment equal to an average household's income over half a year. This clearly illustrates the need for supplementary measures if one wants to assist the rural poor.

### Section E

It would have been easy to criticize the MIRTP for using in the early phases a top-down approach, in which decisions were made by experts and officials before being handed down to communities, but it was necessary to start the process from the level of the governmental authorities of the district. It would have been difficult to respond to the requests of villagers and other rural inhabitants without the support and understanding of district authorities.

### Section F

Today, nobody in the district argues about the importance of improved paths and inexpensive means of transport. But this is the result of dedicated work over a long period particularly from the officers in charge of community development. They played an essential role in raising awareness and interest among the rural communities. The concept of integrated rural transport is now well established in Tanzania, where a major program of rural transport is just about to start. The experiences from Makete will help in this initiative, and Makete District will act as a reference for future work.

**Questions 1-4:** Complete each sentence with the correct ending, **A-J**, below.

1. Construction of footbridges, steps and handrails
  2. Frequent breakdown of buses and trucks in Makete
  3. The improvement of secondary roads and paths
  4. The isolation of Makete for part of the year
- 
- A. Provided the people of Makete with experience in running bus and truck services.
  - B. Was especially successful in the northern part of the district.
  - C. Differed from earlier phases in that the community became less actively involved.
  - D. Improved paths used for transport up and down hillsides.
  - E. Was no longer a problem once the roads had been improved.
  - F. Cost less than locally made wheelbarrows.
  - G. Was done only at the request of local people who were willing to lend a hand.
  - H. Was at first considered by MIRTP to be affordable for the people of the district.
  - I. Hindered attempts to make the existing transport services more efficient.
  - J. Was thought to be the most important objective of Phase III

## Practise Reading Passage 1.9

### ENDLESS HARVEST

More than two hundred years ago, Russian explorers and fur hunters landed on the Aleutian Islands, a volcanic archipelago in the North Pacific, and learned of a land mass that lay farther to the north. The islands' native inhabitants called this land mass Aleyska - the 'Great Land'; today, we know it as Alaska.

The forty-ninth state to join the United States of America (in 1959), Alaska is fully one-fifth the size of the mainland 48 - states combined. It shares, with Canada, the second, longest river system in North America and has over half the coastline of the United States. The rivers feed into the Bering Sea and Gulf of Alaska - cold, nutrient-rich waters which support tens of millions of seabirds, and over 400 species of fish, shellfish, crustaceans, and mollusks. Taking advantage of this rich bounty, Alaska's commercial fisheries have developed into some of the largest in the world.

According to the Alaska Department of Fish and Game (ADF&G), Alaska's commercial fisheries landed hundreds of thousands of tonnes of shellfish and herring, and well over a million tonnes of ground fish (cod, sole, perch and pollock) in 2000. The true cultural heart and soul of Alaska's fisheries, "however, is salmon. 'Salmon,' notes writer Susan Ewing in The Great Alaska Nature Factbook, pump through Alaska like blood through a heart, bringing rhythmic, circulating nourishment to land, animals and people.' The 'predictable abundance of salmon allowed some native cultures to flourish,' and 'dying spankers" feed bears, eagles, other animals, and ultimately the soil itself' All five species of Pacific salmon - chinook, or king; chum, or dog; Coho, or silver; sockeye, or red; and pink, or humpback - spawn in Alaskan waters, and 90% of all Pacific salmon commercially caught in North America are produced there. Indeed, if Alaska was an independent nation, it would be the largest producer of wild salmon in the world. During 2000, commercial catches of Pacific salmon in Alaska exceeded 320,000 tonnes, with an ex-vessel value of over \$US260 million.

Catches have not always been so healthy. Between 1940 and 1959, over-fishing led to crashes in salmon populations so severe that in 1953 Alaska was declared a federal disaster area. With the onset of statehood, however, the State of Alaska took over management of its own fisheries, guided by a state constitution which mandates that Alaska's natural resources be managed on a sustainable basis. At that time, statewide harvests totaled around 25 million salmon. Over the next few- decades average catches steadily increased as a result of this policy of sustainable management, until, during the 1990s, annual harvests were well in excess of 100 million, and on several occasions over 200 million fish.

The primary reason for such increases is what is known as 'In-Season Abundance-Based Management'. There are biologists throughout the state constantly monitoring adult fish as they show up to spawn. The biologists sit in streamside counting towers, study sonar, watch from aeroplanes, and talk to fishermen. The salmon season in Alaska is not pre-set. The fishermen know the approximate time of year when they will be allowed to fish, but on any given day, one or more field biologists in a particular area can put a halt to fishing. Even sport fishing can be brought to a halt. It is this management mechanism that has allowed Alaska salmon stocks - and, accordingly, Alaska salmon fisheries - to prosper, even as salmon populations in the rest of the United States are increasingly considered threatened or even endangered.

In 1999, the Marine Stewardship Council (MSC)\*\*\* commissioned a review of the Alaska salmon fishery. The Council, which was founded in 1996, certifies fisheries that meet high environmental standards, enabling them to use a label that recognises their environmental responsibility. The MSC has established a set of criteria by which commercial fisheries can be judged. Recognising the potential benefits of being identified as environmentally responsible, fisheries approach the Council requesting to undergo the certification process. The MSC then appoints a certification committee, composed of a panel of fisheries experts, which gathers information and opinions from fishermen, biologists, government officials, industry representatives, non-governmental organisations and others.

Some observers thought the Alaska salmon fisheries would not have any chance of certification when, in the months leading up to MSC's final decision, salmon runs throughout western Alaska - completely collapsed. In the Yukon and Kuskokwim rivers, chinook and chum runs were probably the poorest since statehood; subsistence communities throughout the region, who normally have priority over commercial fishing, were devastated.

The crisis was completely unexpected, but researchers believe it had nothing to do with impacts of fisheries. Rather, they contend, it was almost certainly the result of climatic shifts, prompted in part by cumulative effects of the el nino/la nina phenomenon on Pacific Ocean temperatures, culminating in a harsh winter in which huge numbers of salmon eggs were frozen. It could have meant the end as far as the certification process was concerned. However, the state reacted quickly, closing down all fisheries, even those necessary for subsistence purposes.

In September 2000, MSC announced that the Alaska salmon fisheries qualified for certification. Seven companies producing Alaska salmon were immediately granted permission to display the MSC logo on their products. Certification is for an initial period of five years, with an annual review to ensure that the fishery is continuing to meet the required standards.

Questions 1-6

Complete each sentence with the correct ending, **A-K**. below.

Write the correct letter, A-K. in boxes 1-6 on your answer sheet.

1. In Alaska, biologists keep a check on adult fish
2. Biologists have the authority
3. In-Season Abundance-Based Management has allowed the Alaska salmon fisheries
4. The Marine Stewardship Council (MSC) was established
5. As a result of the collapse of the salmon runs in 1999, the state decided
6. In September 2000, the MSC allowed seven Alaska salmon companies

- A. to recognise fisheries that care for the environment.
- B. to be successful.
- C. to stop fish from spawning
- D. to set up environmental protection laws.
- E. to stop people fishing for sport.
- F. to label their products using the MSC logo.
- G. to ensure that fish numbers are sufficient to permit fishing.
- H. to assist the subsistence communities in the region.
- I. to freeze a huge number of salmon eggs.
- J. to deny certification to the Alaska fisheries.
- K. to close down all-fisheries.

## Practise Reading Passage 1.10

### EXPERIENCE VERSUS SPEED

*Certain mental functions slow down with age, but the brain compensates in ways that can keep seniors as sharp as youngsters.*

Jake, aged 16, has a terrific relationship with his grandmother Rita, who is 70. They live close by, and they even take a Spanish class together twice a week at a local college. After class, they sometimes stop at a cafe for a snack. On one occasion, Rita tells Jake, 'I think it's great how fast you pick up new grammar. It takes me a lot longer.' Jake replies, 'Yeah, but you don't seem to make as many silly mistakes on the quizzes as I do. How do you do that?'

In that moment, Rita and Jake stumbled across an interesting set of differences between older and younger minds. Popular psychology says that as people age their brains 'slow down'. The implication, of course, is that elderly men and women are not as mentally agile as middle-aged adults or even teenagers. However, although certain brain functions such as perception and reaction time do indeed take longer, that slowing down does not necessarily undermine mental sharpness. Indeed, evidence shows that older people are just as mentally fit as younger people because their brains compensate for some kinds of declines in creative ways that young minds do not exploit.

Just as people's bodies age at different rates, so do their minds. As adults advance in age, the perception of sights, sounds and smells takes a bit longer, and laying down new information into memory becomes more difficult. The ability to retrieve memories also quickly slides and it is sometimes harder to concentrate and maintain attention.

On the other hand, the ageing brain can create significant benefits by tapping into its extensive hoard of accumulated knowledge and experience. The biggest trick that older brains employ is to use both hemispheres simultaneously to handle tasks for which younger brains rely predominantly on one side. Electronic images taken by cognitive scientists at the University of Michigan, for example, have demonstrated that even when doing basic recognition or memorization exercises, seniors exploit the left and right side of the brain more extensively than men and women who are decades younger. Drawing on both sides of the brain gives them a tactical edge, even if the speed of each hemisphere's process is slower.

In another experiment, Michael Falkenstein of the University of Dortmund in Germany found that when elders were presented with new computer exercises they paused longer before reacting and took longer to complete the tasks, yet they made 50% fewer errors, probably because of their more deliberate pace.

One analogy for these results might be the question of who can type a paragraph 'better': a 16-year-old who glides along at 60 words per minute but has to double back to correct a number of mistakes or a 70-year-old who strikes keys at only 40 words per minute but spends less time

fixing errors? In the end, if 'better' is defined as completing a clean paragraph, both people may end up taking the same amount of time.

Computerized tests support the notion that accuracy can offset speed. In one so-called distraction exercise, subjects were told to look at a screen, wait for an arrow that pointed in a certain direction to appear, and then use a mouse to click on the arrow as soon as it appeared on the screen. Just before the correct symbol appeared, however, the computer displayed numerous other arrows aimed in various other directions. Although younger subjects cut through the confusion faster when the correct arrow suddenly popped up, they more frequently clicked on incorrect arrows in their haste.

Older test takers are equally capable of other tasks that do not depend on speed, such as language comprehension and processing. In these cases, however, the elders utilize the brain's available resources in a different way. Neurologists at Northwest University came to this conclusion after analyzing 50 people ranging from age 23 to 78. The subjects had to lie down in a magnetic resonance imaging (MRI) machine and concentrate on two different lists of printed words posted side by side in front of them. By looking at the lists, they were to find pairs of words that were similar in either meaning or spelling.

The eldest participants did just as well on the tests as the youngest did, and yet the MRI scans indicated that in the elders' brains, the areas which are responsible for language recognition and interpretation were much less active. The researchers did find that the older people had more activity in brain regions responsible for attentiveness. Darren Gleitman, who headed the study, concluded that older brains solved the problems just as effectively but by different means.

### Questions 1-4

Complete each sentence with the correct ending A-F.

Write the correct letter **A-F** in boxes 1-4 on your answer sheet

1. According to popular psychology
  2. Researchers at the University of Michigan showed that
  3. Michael Falkenstein discovered that
  4. Scientists at Northwest University concluded that
- 
- A. the older we get the harder it is to concentrate for any length of time.
  - B. seniors take longer to complete tasks but with greater accuracy.
  - C. old people use both parts of their brain more than young people.
  - D. older people use their brains differently but achieve the same result.
  - E. the speed of our brain decreases with age.
  - F. older people do not cope well with new technology.

## ANSWER KEY

|                                                                |                                                                |                                                                        |
|----------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------|
| <b>PASSAGE 1.1</b><br><br>1. D<br>2. C<br>3. A<br>4. E         | <b>PASSAGE 1.2</b><br><br>1. B<br>2. E<br>3. A<br>4. C<br>5. G | <b>PASSAGE 1.3</b><br><br>1. G<br>2. C<br>3. F<br>4. I<br>5. B         |
| <b>PASSAGE 1.4</b><br><br>1. E<br>2. H<br>3. D<br>4. G<br>5. B | <b>PASSAGE 1.5</b><br><br>1. B<br>2. G<br>3. A<br>4. D<br>5. E | <b>PASSAGE 1.6</b><br><br>1. M<br>2. E<br>3. G<br>4. P<br>5. J<br>6. B |
| <b>PASSAGE 1.7</b><br><br>1. G<br>2. E<br>3. H<br>4. C         | <b>PASSAGE 1.8</b><br><br>1. D<br>2. I<br>3. G<br>4. E         | <b>PASSAGE 1.9</b><br><br>1. G<br>2. E<br>3. B<br>4. A<br>5. K<br>6. F |
| <b>PASSAGE 1.10</b><br><br>1. E<br>2. C<br>3. B<br>4. D        |                                                                |                                                                        |